

EDMONTON STONY PLAIN, AB, Canada

WMO: 711270

Lat: 53.5475N

Lon: 114.1083W

Elev: 766

StdP: 92.45

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-25.9	-32.9	0.2	-28.6	-29.4	0.3	-25.6	9.8	0.0	8.6	-0.5	1.9	270	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	27.2	17.2	25.6	16.4	24.0	15.6	18.8	24.9	17.7	23.5	16.7	22.2	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	13.0	21.5	15.5	12.1	20.5	14.4	11.2	19.3	56.8	25.0	53.0	23.7	49.7	22.3	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.0	6.0	-31.8	30.5	3.6	2.1	-34.4	32.0	-36.5	33.2	-38.5	34.4	-41.1	35.9
			-32.3	20.7	3.2	1.4	-34.6	21.7	-36.5	22.5	-38.2	23.3	-40.5	24.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-9.7	-7.9	-3.4	4.3	10.7	14.6	17.2	16.0	11.3	4.7	-3.5	-9.2
	DBStd	11.47	9.11	7.85	7.53	5.35	4.57	3.16	3.03	3.32	4.40	4.86	6.82	8.15
	HDD10.0	2968	611	501	417	181	46	3	0	2	36	173	404	595
	HDD18.3	5342	869	734	675	422	237	119	58	86	212	422	654	853
	CDD10.0	714	0	0	0	9	69	141	223	188	76	9	0	0
	CDD18.3	46	0	0	0	0	2	6	23	13	2	0	0	0
	CDH23.3	486	0	0	0	1	34	68	201	139	42	2	0	0
	CDH26.7	76	0	0	0	0	3	9	35	23	6	0	0	0
Wind	WSAvg	2.6	2.6	2.6	2.7	3.0	2.9	2.5	2.4	2.3	2.5	2.8	2.6	2.8
Precipitation	PrecAvg	445	21	13	17	31	50	78	89	55	35	23	19	14
	PrecMax	606	66	34	34	59	113	174	190	119	92	83	56	45
	PrecMin	261	0	0	2	4	11	17	25	27	4	4	2	0
	PrecStd	85	16	8	8	16	25	38	38	20	23	18	14	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	9.4	13.3	22.0	26.8	27.8	29.9	29.1	27.3	21.4	13.1	9.0
		MCWB	3.2	3.4	6.0	10.6	13.6	16.8	18.6	18.0	16.1	11.8	6.5	3.6
	2%	DB	6.3	6.2	9.9	18.5	24.0	25.2	27.3	26.6	24.1	16.9	9.0	5.9
		MCWB	2.3	1.7	3.9	8.9	12.5	15.1	18.2	17.5	14.7	9.7	4.2	1.9
	5%	DB	4.2	4.2	7.7	15.5	21.8	23.2	25.5	24.7	21.7	14.1	6.4	3.8
		MCWB	1.0	0.4	2.7	7.1	11.7	14.3	17.4	16.6	13.6	7.8	2.7	0.6
	10%	DB	2.1	2.2	5.6	12.7	19.2	21.3	23.8	22.9	19.1	11.8	4.4	1.6
		MCWB	-0.4	-0.8	1.6	5.7	10.4	13.4	16.4	15.7	12.3	6.5	1.4	-1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.9	6.3	11.1	15.0	18.1	20.7	19.9	16.8	12.3	6.9	4.2
		MCDB	7.5	8.5	12.5	21.0	23.6	24.3	26.1	26.4	25.2	20.5	12.3	8.5
	2%	WB	2.5	2.0	4.4	9.2	13.4	16.4	19.4	18.5	15.3	10.0	4.8	2.2
		MCDB	5.9	5.6	9.3	17.7	21.7	22.9	25.3	24.7	22.7	15.9	8.4	5.3
	5%	WB	1.2	0.6	3.0	7.7	12.2	15.2	18.2	17.4	14.0	8.4	3.2	0.7
		MCDB	4.0	3.9	7.1	14.6	20.0	21.1	24.0	23.2	20.4	13.2	6.2	3.6
	10%	WB	-0.3	-0.6	1.8	6.1	11.0	14.2	17.2	16.4	12.8	7.1	1.7	-0.9
		MCDB	2.1	2.2	5.3	11.7	18.1	19.7	22.6	21.8	18.5	11.0	4.0	1.5
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.9	8.8	10.2	11.1	10.1	10.4	10.7	10.7	9.1	7.3	7.5
		MCDBR	9.3	10.8	11.3	14.8	14.8	13.2	12.9	13.7	15.3	13.3	9.5	8.5
		MCWBR	6.7	7.3	6.7	7.8	6.7	5.8	5.9	6.2	7.2	7.1	6.1	6.1
	5% WB	MCDBR	9.4	10.6	10.9	14.2	13.5	11.7	12.1	12.8	14.3	12.6	9.3	8.9
		MCWBR	7.2	7.4	6.6	7.8	6.3	5.6	6.1	6.5	7.0	7.1	6.3	6.8
Clear-Sky Solar Irradiance	taub	0.249	0.262	0.298	0.327	0.342	0.350	0.353	0.358	0.309	0.281	0.253	0.238	
	taud	2.333	2.370	2.364	2.389	2.390	2.383	2.425	2.361	2.499	2.557	2.462	2.319	
	Ebn at Noon	749	860	892	906	906	899	889	862	874	835	751	689	
	Edh at Noon	55	74	93	105	111	113	106	107	82	62	49	46	
All-Sky Solar Radiation	RadAvg	0.96	1.99	3.49	4.65	5.63	5.97	6.09	5.03	3.48	2.00	0.98	0.70	
	RadStd	0.09	0.15	0.24	0.43	0.41	0.40	0.34	0.34	0.36	0.22	0.10	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.63	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page